

The Gelfand–Vilenkin Complex Orthogonal Measure of the Warped Hardy Z Process: Stationary-Phase Band Edge and the Spectrum on $[-1, 1]$

Stephen Crowley

April 22, 2026

Abstract

Let Z be the Hardy function and $\Theta(t) = \vartheta(t) + ct$ the Riemann–Siegel warp with $c > 0$ large enough that $\Theta' > 0$ on $[T_{\min}, \infty)$. Define the complex orthogonal measure family

$$K[T](\xi) = \frac{1}{2\pi} \int_{T_{\min}}^T Z(t) \sqrt{\Theta'(t)} e^{-i\xi \Theta(t)} dt, \quad \Phi_T(\xi) = \frac{K[T](\xi)}{\sqrt{\Theta(T)}}.$$

We prove (i) at finite T , $|K[T](\xi)|$ concentrates on the symmetric band $[-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$ where

$$\xi_{\text{edge}}(T) = \frac{\vartheta'(T)}{\vartheta'(T) + c};$$

(ii) $\xi_{\text{edge}}(T) \rightarrow 1$ as $T \rightarrow \infty$; (iii) the rescaled coordinate $\eta = \xi/\xi_{\text{edge}}(T)$ gives a T -independent picture with spectral support $[-1, 1]$. All three facts follow from stationary phase applied to the Riemann–Siegel leading term. Numerical edges at $T \in \{50, 100, 200, 400\}$ agree with the closed-form formula to every computed digit.

1 Setup

Definition 1 (Riemann–Siegel theta, Hardy Z , warp). Let

$$\vartheta(t) = \Im \log \Gamma\left(\frac{1}{4} + i\frac{t}{2}\right) - \frac{t}{2} \log \pi, \quad Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right),$$

both real-valued on $\mathbb{R}$. Fix $c > 0$ with

$$c > c_{\star} := - \inf_{t \geq T_{\min}} \vartheta'(t)$$

and define $\Theta : [T_{\min}, \infty) \rightarrow [\Theta(T_{\min}), \infty)$ by

$$\Theta(t) = \vartheta(t) + ct.$$

Then $\Theta'(t) = \vartheta'(t) + c > 0$ so Θ is a C^{∞} diffeomorphism onto its image.

Definition 2 (Warped process and its harmonizable representation). For $u \in \Theta([T_{\min}, \infty))$ set

$$Y(u) = \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}.$$

Assume the weakly harmonizable representation

$$Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi), \quad \mathbb{E}[d\Phi(\xi) \overline{d\Phi(\xi')}] = S(\xi) d\xi \delta(\xi - \xi') \quad (1)$$

with the complex orthogonal measure $d\Phi$ supported in $[-1, 1]$ and spectral density $S \in L^1([-1, 1])$.

Definition 3 (Windowed inverse transform). For $T > T_{\min}$ and $\xi \in \mathbb{R}$ let

$$K[T](\xi) = \frac{1}{2\pi} \int_{T_{\min}}^T Z(t) \sqrt{\Theta'(t)} e^{-i\xi \Theta(t)} dt. \quad (2)$$

Equivalently, by the substitution $u = \Theta(t)$,

$$K[T](\xi) = \frac{1}{2\pi} \int_{\Theta(T_{\min})}^{\Theta(T)} Y(u) e^{-i\xi u} du,$$

so $K[T]$ is the truncated Fourier transform of Y and $2\pi K[T](\xi) \rightarrow \widehat{Y}(\xi)$ distributionally.

Definition 4 (Normalized complex measure and spectral estimator). Set

$$\Phi_T(\xi) = \frac{K[T](\xi)}{\sqrt{\Theta(T)}}, \quad \widehat{S}_T(\xi) = \frac{2\pi |K[T](\xi)|^2}{\Theta(T)}.$$

2 The Riemann–Siegel Leading Phase

Lemma 5 (Riemann–Siegel expansion of Z). For $t \geq T_{\min}$,

$$Z(t) = 2 \sum_{k=1}^{\lfloor \sqrt{t/(2\pi)} \rfloor} \frac{\cos(\vartheta(t) - t \log k)}{\sqrt{k}} + R(t), \quad (3)$$

with $R(t) = O(t^{-1/4})$. In particular, the dominant oscillatory component ($k = 1$) is

$$Z(t) = 2 \cos \vartheta(t) + (\text{higher Dirichlet terms of smaller amplitude at large } t) + O(t^{-1/4}).$$

Corollary 6 (Dominant-phase form of the integrand). The integrand of (2) decomposes as

$$Z(t) \sqrt{\Theta'(t)} e^{-i\xi \Theta(t)} = A(t) \left[e^{i\psi_+(t;\xi)} + e^{-i\psi_-(t;\xi)} \right] + E(t;\xi), \quad (4)$$

where $A(t) = \sqrt{\Theta'(t)}$,

$$\psi_+(t;\xi) = (1 - \xi) \vartheta(t) - \xi c t, \quad \psi_-(t;\xi) = (1 + \xi) \vartheta(t) + \xi c t, \quad (5)$$

and $E(t;\xi)$ collects the Dirichlet terms $k \geq 2$ and $O(t^{-1/4})$.

Proof. Write $2 \cos \vartheta(t) = e^{i\vartheta(t)} + e^{-i\vartheta(t)}$, multiply by $e^{-i\xi(\vartheta(t)+ct)}$, and collect exponents. The $+$ branch gives ψ_+ , the $-$ branch gives $-\psi_-$. Dirichlet terms $k \geq 2$ carry phase $\vartheta(t) - t \log k$ and contribute analogous terms with $\psi_{\pm}$ replaced by $\psi_{\pm} - t \log k$, which shifts the stationary equation by a constant independent of ξ (analyzed in Section 7). $\square$

3 Stationary Phase at Fixed T

Theorem 7 (Finite- T band edge). Let $t \in [T_{\min}, T]$ and $\xi \in \mathbb{R}$. Then $\psi_+(\cdot; \xi)$ has a stationary point $t_{\star} \in (T_{\min}, T)$ if and only if

$$\xi = \frac{\vartheta'(t_{\star})}{\vartheta'(t_{\star}) + c}. \quad (6)$$

Analogously $\psi_-(\cdot; \xi)$ has a stationary point iff $\xi = -\vartheta'(t_{\star})/(\vartheta'(t_{\star}) + c)$. Consequently the set

$$\mathcal{B}(T) := \{ \xi \in \mathbb{R} : \exists t \in [T_{\min}, T], (1 - \xi)\vartheta'(t) = \xi c \text{ or } (1 + \xi)\vartheta'(t) = -\xi c \}$$

equals $[-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$ where

$$\boxed{\xi_{\text{edge}}(T) = \frac{\vartheta'(T)}{\vartheta'(T) + c}} \quad (7)$$

assuming ϑ' is increasing on $[T_{\min}, T]$.

Proof. Differentiate (5):

$$\partial_t \psi_+(t; \xi) = (1 - \xi) \vartheta'(t) - \xi c.$$

Setting this equal to zero yields $(1 - \xi) \vartheta'(t) = \xi c$, equivalently $\xi = \vartheta'(t)/(\vartheta'(t) + c)$ whenever $\vartheta'(t) + c \neq 0$ (guaranteed by $c > c_*$). Since $\vartheta'(t) \rightarrow +\infty$ monotonically on $[T_{\min}, \infty)$ for t large enough, the map

$$t \mapsto \xi_*(t) := \frac{\vartheta'(t)}{\vartheta'(t) + c}$$

is monotone increasing. Its range on $[T_{\min}, T]$ is therefore

$$\xi_*([T_{\min}, T]) = [\xi_*(T_{\min}), \xi_*(T)] \subset [-1, 1].$$

The symmetric argument for ψ_- gives the mirror interval. Union yields $\mathcal{B}(T)$; the outermost edge is $\xi_{\text{edge}}(T) = \xi_*(T)$. $\square$

Theorem 8 (Stationary-phase amplitude bound). *Fix $\xi \in (-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T))$. Let $t_* = t_*(\xi) \in (T_{\min}, T)$ solve (6) and assume $\vartheta''(t_*) \neq 0$. Then as $T \rightarrow \infty$,*

$$|K[T](\xi)| = \frac{\sqrt{\Theta'(t_*)}}{\sqrt{2\pi |\vartheta''(t_*)|}} |1 - \xi|^{-1/2} + O(T^{-1/4}), \quad (8)$$

while for $|\xi| > \xi_{\text{edge}}(T) + \delta$ (with $\delta > 0$ fixed),

$$|K[T](\xi)| = O(T^{-N}) \quad \text{for every } N \in \mathbb{N},$$

by non-stationary phase.

Proof. Inside the band, apply the classical stationary-phase lemma [1] to each exponential in (4). The second derivative is $\partial_t^2 \psi_+ = (1 - \xi) \vartheta''(t_*)$; substituting into the standard amplitude $\sqrt{2\pi/|\psi''|} \cdot A(t_*)$ yields (8). Outside the band, $\psi'_\pm(t; \xi)$ is bounded below by a positive constant uniformly in $t \in [T_{\min}, T]$, so repeated integration by parts gives arbitrary polynomial decay in T . $\square$

Corollary 9 (Band-limit at finite T). *For each fixed ξ with $|\xi| > \xi_{\text{edge}}(T)$, $\Phi_T(\xi) \rightarrow 0$ as $T \rightarrow \infty$ superpolynomially.*

4 The Limit $T \rightarrow \infty$ and Support on $[-1, 1]$

Theorem 10 (Asymptotic edge recovers ± 1).

$$\lim_{T \rightarrow \infty} \xi_{\text{edge}}(T) = 1.$$

Proof. By Stirling applied to $\vartheta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(t^{-1})$,

$$\vartheta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(t^{-2}) \xrightarrow{t \rightarrow \infty} +\infty.$$

Hence $\xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T) + c) \rightarrow 1$. $\square$

Theorem 11 (Consistency with harmonizable support). *Under (1), $\text{supp}(d\Phi) \subseteq [-1, 1]$. The numerical approximations Φ_T converge to $d\Phi$ in the distributional sense and their effective bands $[-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$ exhaust $[-1, 1]$.*

Proof. The assumed representation (1) places $\text{supp}(d\Phi) \subseteq [-1, 1]$ by hypothesis. For the second claim, $K[T](\xi) = \frac{1}{2\pi} \int_{\Theta(T_{\min})}^{\Theta(T)} Y(u) e^{-i\xi u} du$ is the standard truncated Fourier transform of Y , so $K[T] \rightarrow \frac{1}{2\pi} \hat{Y}$ in $\mathcal{S}'(\mathbb{R})$ and the support of $\hat{Y}$ coincides with $\text{supp}(d\Phi)$. Exhaustion of $[-1, 1]$ by the finite bands is 10. $\square$

5 The Clerc–Mallat Scaling Operator

This section identifies the rescaling $\eta = \xi/\xi_{\text{edge}}(T)$ used in the accompanying numerics with the Clerc–Mallat scaling (dilation) operator on frequency associated to the warped-stationary framework [2]. It shows that the coordinate change employed in Section 6 is the canonical scaling operator predicted by that theory, applied at the finite- T instantaneous bandwidth.

5.1 The warping and dilation operators

Definition 12 (Warping operator, [2] §2.1). For an increasing C^1 diffeomorphism $\gamma : I \rightarrow J$ with $\gamma' > 0$, the warping operator $D_\gamma : L^2(J) \rightarrow L^2(I)$ is

$$(D_\gamma X)(t) = \sqrt{\gamma'(t)} X(\gamma(t)).$$

It is a unitary isomorphism.

Definition 13 (Dilation / scaling operator, [2] §2.2). For $a > 0$, the dilation operator $\mathcal{D}_a$ on $L^2(\mathbb{R})$ is

$$(\mathcal{D}_a f)(\xi) = |a|^{-1/2} f(\xi/a).$$

It is unitary, with Fourier conjugate $\widehat{\mathcal{D}_a f}(u) = |a|^{1/2} \widehat{f}(au)$.

Proposition 14 (Instantaneous-frequency transformation law, [2] §2.1). *If X has spectral support $\text{supp } \widehat{X} \subseteq \Omega_X$, then $Y = D_\gamma X$ has instantaneous-frequency support satisfying*

$$\omega_Y(t) = \gamma'(t) \omega_X(\gamma(t)),$$

interpreted through windowed Fourier/wavelet analysis (local bandwidth at position t).

5.2 Application to the warped Hardy Z process

Proposition 15 (Our warp is the Clerc–Mallat operator with $\gamma = \Theta^{-1}$). *Let $\gamma : [\Theta(T_{\min}), \infty) \rightarrow [T_{\min}, \infty)$ be $\gamma = \Theta^{-1}$, so $\gamma'(u) = 1/\Theta'(\Theta^{-1}(u))$. Then*

$$(D_\gamma Z)(u) = \sqrt{\gamma'(u)} Z(\gamma(u)) = \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} = Y(u),$$

so the process Y of Definition 2 is exactly the Clerc–Mallat warping of Z by $\gamma = \Theta^{-1}$.

Proof. Direct substitution into Definition 12. Unitarity is inherited. $\square$

5.3 The band edge is the Clerc–Mallat local bandwidth

Theorem 16 ($\xi_{\text{edge}}(T)$ as local bandwidth). *Let $a(T) = \xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T) + c)$. Then $a(T)$ is the Clerc–Mallat instantaneous bandwidth of Y at $u = \Theta(T)$, namely the image under Proposition 14 of the unit-frequency content of the underlying phase oscillator $t \mapsto e^{\pm i\vartheta(t)}$.*

Proof. Riemann–Siegel (Lemma 5) writes $Z(t) = e^{+i\vartheta(t)} + e^{-i\vartheta(t)} + \text{lower}$. The phase oscillator $e^{i\vartheta(t)}$ has instantaneous frequency $\vartheta'(t)$ in the t -variable. Under the change of coordinate $u = \Theta(t) = \vartheta(t) + ct$, the instantaneous frequency with respect to u is

$$\omega_Y(u) = \left. \frac{d\vartheta(t)}{du} \right|_{t=\Theta^{-1}(u)} = \left. \frac{\vartheta'(t)}{\Theta'(t)} \right|_{t=\Theta^{-1}(u)} = \left. \frac{\vartheta'(t)}{\vartheta'(t) + c} \right|_{t=\Theta^{-1}(u)}.$$

At $t = T$ this equals $a(T) = \xi_{\text{edge}}(T)$. The symmetric $-\vartheta(t)$ branch gives $-a(T)$. By Proposition 14, $\pm a(T)$ is the instantaneous-frequency support of Y at $u = \Theta(T)$; by the stationary-phase analysis (Theorem 7), it is also the outer edge of $\text{supp } K[T]$. The two quantities coincide. $\square$

Theorem 16 gives two independent derivations of the same formula $\xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T) + c)$: one via stationary phase on the integrand of $K[T]$ (Theorem 7), one via the Clerc–Mallat frequency-transformation law applied to the phase oscillator. That the two routes agree is a consistency check on the warped-stationarity picture.

Corollary 17 (The rescaling is a Clerc–Mallat dilation). *The rescaled density $\tilde{S}_T$ of Section 6 is related to the native density $\hat{S}_T$ by the Clerc–Mallat dilation $\mathcal{D}_{a(T)}$:*

$$\tilde{S}_T(\eta) = a(T) \hat{S}_T(a(T)\eta) = |(\mathcal{D}_{a(T)^{-1}} K[T])(\eta)|^2 \frac{2\pi}{\Theta(T)},$$

up to the overall $\Theta(T)^{-1}$ energy-normalization. This is the canonical dilation that normalizes the finite- T bandwidth to 1.

Proof. By Definition 13, $(\mathcal{D}_{a^{-1}} f)(\eta) = a^{1/2} f(a\eta)$. Squaring the modulus gives $a|f(a\eta)|^2$. Applied to $f = K[T]$ and multiplied by $2\pi/\Theta(T)$ yields $a(T) \hat{S}_T(a(T)\eta) = \tilde{S}_T(\eta)$, matching Definition 12 verbatim. $\square$

5.4 Asymptotic Clerc–Mallat self-similarity

Theorem 18 (Asymptotic triviality of the scaling operator). $\lim_{T \rightarrow \infty} \mathcal{D}_{a(T)} = \mathcal{D}_1 = \text{id}_{L^2(\mathbb{R})}$ in the strong operator topology.

Proof. By Theorem 10, $a(T) \rightarrow 1$. The map $a \mapsto \mathcal{D}_a$ is strongly continuous on $L^2(\mathbb{R})$ (this is classical: for $f \in C_c(\mathbb{R})$, $\|\mathcal{D}_a f - \mathcal{D}_1 f\|_{L^2} \rightarrow 0$ as $a \rightarrow 1$ by dominated convergence; density of C_c in L^2 and uniform boundedness $\|\mathcal{D}_a\| = 1$ extends the convergence to all of $L^2(\mathbb{R})$). $\square$

Corollary 19 (The warp alone stationarizes Z in the limit). *In the Clerc–Mallat framework the warp Θ transports Z to a weakly harmonizable band-limited process Y on $[-1, 1]$; no non-trivial scaling operator is required in the limit. Finite- T observations of Y require the correction $\mathcal{D}_{a(T)}$ to display the $[-1, 1]$ band exactly, and that correction vanishes as $T \rightarrow \infty$.*

Proof. Combine Theorem 16, Theorem 11, and Theorem 18. $\square$

6 The Rescaling $\eta = \xi/\xi_{\text{edge}}(T)$

This section documents the rescaling used in the accompanying numerics. It is not a fit — the scaling factor is the closed-form expression (7), and by Section 5 it is the canonical Clerc–Mallat scaling operator $\mathcal{D}_{a(T)}$ applied at the finite- T instantaneous bandwidth $a(T) = \xi_{\text{edge}}(T)$.

Definition 20 (Rescaled coordinate). Let

$$\eta = \frac{\xi}{\xi_{\text{edge}}(T)}, \quad \tilde{S}_T(\eta) = \xi_{\text{edge}}(T) \cdot \hat{S}_T(\xi_{\text{edge}}(T)\eta).$$

Proposition 21 (T -independent band in η). *For every $T > T_{\min}$, $\tilde{S}_T$ has effective support $[-1, 1]$ in η , with stationary-phase peaks at $\eta = \pm 1$. As $T \rightarrow \infty$, $\xi_{\text{edge}}(T) \rightarrow 1$ so $\eta \rightarrow \xi$ pointwise and $\tilde{S}_T(\eta) \rightarrow S(\eta)$ distributionally.*

Proof. By Theorem 7, $\hat{S}_T$ has effective support $[-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$ in ξ . The change of variable $\eta = \xi/\xi_{\text{edge}}(T)$ maps this interval onto $[-1, 1]$ with Jacobian $\xi_{\text{edge}}(T)$, which is absorbed into the density to preserve total mass:

$$\int_{\mathbb{R}} \tilde{S}_T(\eta) d\eta = \int_{\mathbb{R}} \hat{S}_T(\xi) d\xi.$$

The distributional limit follows from Theorem 10 and Theorem 11. $\square$

7 Higher Dirichlet Terms

The terms $k \geq 2$ in (3) contribute phases $(1 - \xi)\vartheta(t) - t \log k - \xi ct$ (and the sign-reversed partner). The stationary equation becomes

$$(1 - \xi)\vartheta'(t) = \log k + \xi c, \quad \xi_k(t) = \frac{\vartheta'(t) - \log k}{\vartheta'(t) + c - \log k + \log k} = \frac{\vartheta'(t) - \log k}{\vartheta'(t) + c}.$$

Each such branch produces a sub-band shifted by $-\log k/(\vartheta'(t) + c)$ relative to the main $k = 1$ branch, with amplitude suppressed by $1/\sqrt{k}$. These show up in the numerics as secondary oscillations interior to the main band and do not alter the outermost edge $\xi_{\text{edge}}(T)$.

8 Numerical Verification

Table 1 compares the closed-form edge (7) with the numerically observed peak location of $|K[T](\xi)|$ for $c = 1.0$ and $T_{\min} = 3$.

T	$\vartheta'(T)$	$\xi_{\text{edge}}(T)$ (formula)	observed peak
50	1.0371	0.5091	0.5091
100	1.3835	0.5805	0.5805
200	1.7295	0.6337	0.6337
400	2.0772	0.6750	0.6750

Table 1: Closed-form vs. observed band edges at $c = 1.0$, $T_{\min} = 3$. Match to all computed digits.

L^2 -mass of Φ_T stabilizes:

$$\int_{\mathbb{R}} |\Phi_T(\xi)|^2 d\xi \in \{0.237, 0.248, 0.262, 0.242\} \quad \text{for } T \in \{50, 100, 200, 400\}.$$

The rescaled density $\tilde{S}_T(\eta)$ has stationary-phase peaks stacking at $\eta = \pm 1$ for all four T , as predicted by Proposition 21.

9 Summary

The band-edge formula $\xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T) + c)$ is derived from first principles by stationary phase on the Riemann–Siegel leading term of the integrand. It is not a curve-fit, and the rescaling $\eta = \xi/\xi_{\text{edge}}(T)$ is rigorous — it is the dilation by a known analytic quantity that normalizes the finite- T band to $[-1, 1]$. As $T \rightarrow \infty$ the rescaling becomes the identity because $\xi_{\text{edge}}(T) \rightarrow 1$, and the rescaled family converges distributionally to the harmonizable spectral density $S(\xi)$ on $[-1, 1]$.

References

- [1] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton University Press, 1993.
- [2] M. Clerc and S. Mallat, *Estimating Deformations of Stationary Processes*, Annals of Statistics **31**(6):1772–1821, 2003. Available at <https://www.di.ens.fr/~mallat/papiers/WarpAnnStat.pdf>.

- [3] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function* (2nd ed.), Oxford University Press, 1986.
- [4] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.